

Centre for Innovation and Entrepreneurship

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MICRO PROJECT ABSTRACT

Title of Idea:	DIY Emergency power box		
Theme:	DIY Emergency Power Box that works as both a USB mobile charger and variable voltage power supply		
Class & Section:	CSE-D		
Components Used:	XL6009 Boost Converter Module – 1 pc 18650 Li-ion Batteries – 2 pcs 18650 Battery Holder (2-cell) – 1 pcs USB Power Bank Module – 1 pc Mini Digital Voltmeter (0.28”) – 1 pc LED strip		
Idea description	The DIY emergency power box provides both 5V USB and adjustable DC output up to 30V, unlike a normal power bank. It can power USB devices like smartphones and DC devices like Arduino, LEDs, and sensors. The XL6009 boosts battery voltage, while the USB module raises 3.7V to 5V. Parallel-connected 18650 batteries increase capacity, and a voltmeter shows the output.		
Week-wise plan of action	STEP 1: Mount the 18650 battery holder, USB power bank module, and XL6009 boost converter on the acrylic sheet. STEP 2: Connect the battery terminals to the USB module, XL6009, and mini voltmeter. STEP 3: Add an ON/OFF switch then wire the voltmeter to the XL6009 output. STEP 4: Insert the batteries, close the acrylic cover, and the DIY emergency power box is ready.		
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